

Constants & Assumptions

Gravitational Constant	9.81
Maximum Landing Weight (MLW)	251,290.00
Main Gear Load Share %	90

Wheel Configuration

No. of Gear Legs	2.00
Wheels per Main Leg	6.00
Factor of Safety	2.50
Bogie-Beam Load Sharing	0.17

FORCE LOAD VALUES

Vertical Load Limit	2.50
Braking Coefficient of Friction	0.80
Side Load Factor (Crosswind)	0.50
Boeing Maximum Tyre Load	26,980.00

*Factor of Safety will be performed in Fusion 360 Simulation Analysis
Assumes equal distribution of forces to all axel stubs of bogie-beam.
Maximum Value of Mass applied to tyres from BOEING safety standards*

STATIC VALUES

Total Aircraft Weight	2,465,154.90
Vertical Load on main gear	1,232,577.45
Vertical Load per wheel	205,429.58

Main Gear Wheel Loads

Vertical F _z	513,573.94
Braking F _x	410,859.15
Crosswind F _y	102,714.79
Vertical F _z Ultimate Limit (FoS)	1,283,934.84
Braking F _x Ultimate Limit (FoS)	1,027,147.88
Crosswind F _y Ultimate Limit (FoS)	256,786.97

Total Gear Wheel Load

Vertical F _z	3,081,443.63
Braking F _x	2,465,154.90
Crosswind F _y	616,288.73
Vertical F _z Ultimate Limit (FoS)	7,703,609.06
Braking F _x Ultimate Limit (FoS)	6,162,887.25
Crosswind F _y Ultimate Limit (FoS)	1,540,721.81

Minimum Safety Check

264673.8

Calculated F_z Limit as % of Tyre Load

194.0403385

// Assumed Load is LARGER than Tyre Rated Load. This means Vertical Load Limit

// is an overestimate to force experience during landing

